

CPU Specification

1. Global Interface

Signal	Direction	Width	Description
clk	Input	1	System clock (posedge triggered)
rst_n	Input	1	Asynchronous active-low reset (0 = reset)
prog_nibble_in	Input	4	Serial command/data input (4-bit nibble)
prog_nibble_out	Output	4	Serial result output (4-bit nibble)
prog_out_valid	Output	1	Asserted when prog_nibble_out contains valid data
prog_done	Output	1	Operation completion indicator • Success: 1-cycle pulse • Error: 2-cycle pulse
mem_addr	Output	12	Memory address (0-4095)
mem_wdata	Output	32	Memory write data
mem_rdata	Input	32	Memory read data
mem_wen	Output	1	Write enable (1 = write, 0 = read)
mem_cen	Output	1	Chip enable (1 = memory access active)

2. Programming Frame Format (Serial Protocol)

- **Transmission rate:** 4 bits (1 nibble) per clock cycle
- **Frame length:** 13 nibbles (52 bits total)
- **Transmission order:** LSB-first for each field, fields transmitted in order below:

Field	Bits	Nibbles	Description	Transmission Order
Address	12	3	Target address [11:0]	LSB nibble first: addr[3:0], addr[7:4], addr[11:8]
Command	8	2	Operation code [7:0]	cmd[3:0], cmd[7:4]
Write Data	32	8	Data for write operations [31:0]	wdata[3:0], wdata[7:4], ..., wdata[31:28]

Example: For address 12'h123, command 8'h45, data 32'h89ABCDEF:

- Nibble 0: 3 (addr[3:0])
- Nibble 1: 2 (addr[7:4])
- Nibble 2: 1 (addr[11:8])
- Nibble 3: 5 (cmd[3:0])
- Nibble 4: 4 (cmd[7:4])
- Nibble 5: F (wdata[3:0])
- Nibble 6: E (wdata[7:4])
- Nibble 7: D (wdata[11:8])
- Nibble 8: C (wdata[15:12])
- Nibble 9: B (wdata[19:16])
- Nibble 10: A (wdata[23:20])
- Nibble 11: 9 (wdata[27:24])
- Nibble 12: 8 (wdata[31:28])

3. Command Set

3.1. Basic Memory Operations

Code (hex)	Mnemonic	Description
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Code (hex)	Mnemonic	Description
0x00	FILL0	Write 0 from address A to 4095
0x01	FILL1	Write 1 from address A to 4095
0x02	WRITE	Write wdata to address A
0x03	READ	Read from address A (output on nibble_out)
0x04	WRITE_INC	Write wdata to A, then inc data & addr until 4095
0x05	READ_SEQ	Read from A to 4095, output each word
0x06	INC_ACC	Read A, increment, write back to A
0x07-0x0F	RESERVED	No operation, immediate done

3.2. Arithmetic & Logical Operations

All operations use 3 consecutive addresses unless specified

Code	Mnemonic	Description
0x10	ADD	$R[A+2] = R[A] + R[A+1]$
0x11	SUB	$R[A+2] = R[A] - R[A+1]$
0x12	MUL	$\{R[A+3], R[A+2]\} = R[A] * R[A+1]$
0x13	DIV	$R[A+2] = \text{quotient}, R[A+3] = \text{remainder}$

Code	Mnemonic	Description
0x14	INV	$R[A+1] = \sim R[A]$
0x15	AND	$R[A+2] = R[A] \& R[A+1]$
0x16	OR	$R[A+2] = R[A] \mid R[A+1]$
0x17	XOR	$R[A+2] = R[A] \wedge R[A+1]$

3.4. Reserved Commands

- **0x18-0xFF:** No operation, immediate `prog_done` (1 cycle)
- **Implementation note:** Can be extended with new commands

4. Output Serialization

- **32-bit read data** serialized as 8 nibbles (4 bits each)
- **Transmission order:** LSB nibble first (`wdata[3:0]` to `wdata[31:28]`)
- `prog_out_valid` asserted for each valid nibble
- **READ command:** Outputs 1 word (8 nibbles)
- **READ_SEQ command:** Outputs each word sequentially (8 nibbles per word)

5. Error Handling

Error Condition	<code>prog_done</code> Behavior	Notes
Address out of range	2-cycle pulse	Any address > 4095
Arithmetic overflow	1-cycle pulse (no error)	Result truncated to 32 bits
Division by zero	2-cycle pulse	For DIV command only

Error Condition	prog_done Behavior	Notes
Invalid command	1-cycle pulse (NOP)	Reserved commands treated as NOP

6. Reset Behavior

- All internal registers cleared
- prog_done = 0
- prog_out_valid = 0
- Memory contents preserved (memory is separate)
- Ready to receive new frame

Timing diagram examples



